Hybrid Variable Elimination Ordering: Experimental Results

TTIC 31180 — Probabilistic Graphical Models Final Project

Setup

We evaluated min-degree, min-fill, weighted min-fill, and the proposed hybrid heuristic across five topology families. The experiment used 71 graph instances and 715 successful method-instance runs. Explicit pairwise variable elimination was attempted only when the projected intermediate factor stayed below the configured cutoff; 373 of 715 successful structural runs completed explicit VE under this cutoff. The moralized Bayesian-network group includes a manually encoded Asia network and built-in BN-like layered DAGs because the local environment did not include `pgmpy` benchmark loaders.

Topology	Instances	Median n	Max n	Median density	Median max degree
Low-treewidth	8	75	200	0.032	5.0
Random	27	80	120	0.064	9.0
Grid	4	82	144	0.046	4.0
Scale-free	27	100	150	0.040	22.0
Moralized BN	5	33	58	0.149	12.0

Structural Results

The primary metrics are structural: induced width and peak factor size. Lower is better for both. The following tables report medians within each topology.

Topology	Min-degree	Min-fill	Weighted	Hybrid
Low-treewidth	1.0	1.0	1.0	1.0
Random	22.0	20.0	21.0	21.0
Grid	11.5	11.5	11.5	11.0
Scale-free	10.0	10.0	10.0	10.0
Moralized BN	7.0	7.0	7.0	7.0

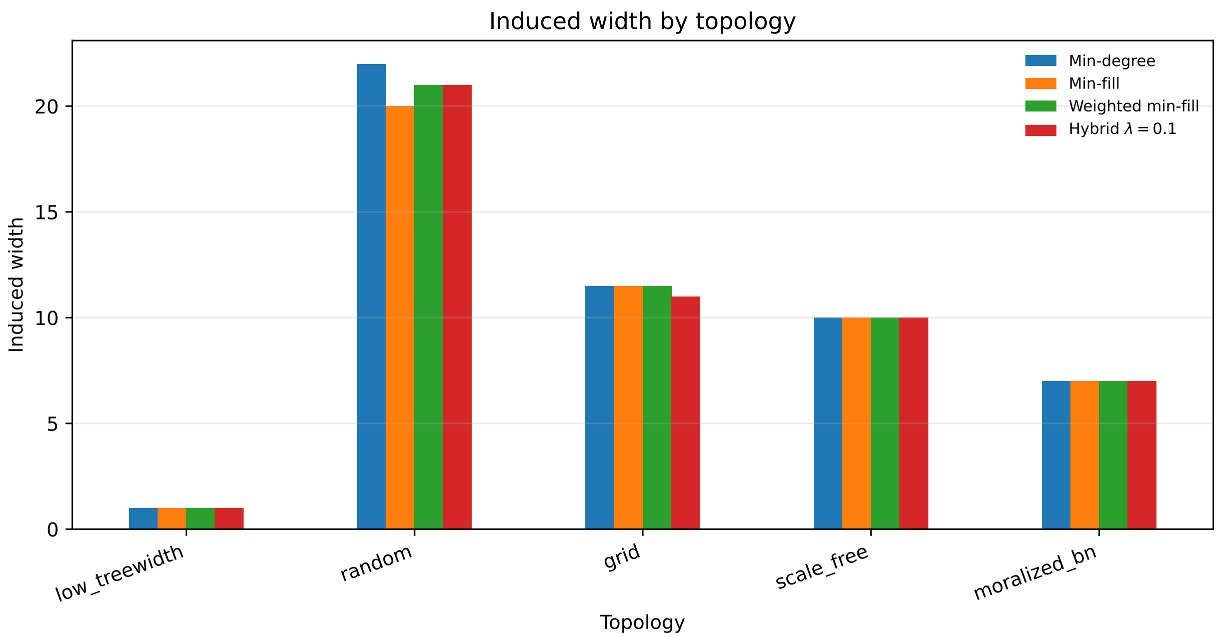
Topology	Min-degree	Min-fill	Weighted	Hybrid
Low-treewidth	1.40	1.40	1.40	1.40
Random	10.10	10.14	10.07	10.20
Grid	5.90	5.83	5.08	4.95
Scale-free	6.72	6.99	6.72	7.22
Moralized BN	3.54	3.36	3.36	3.36

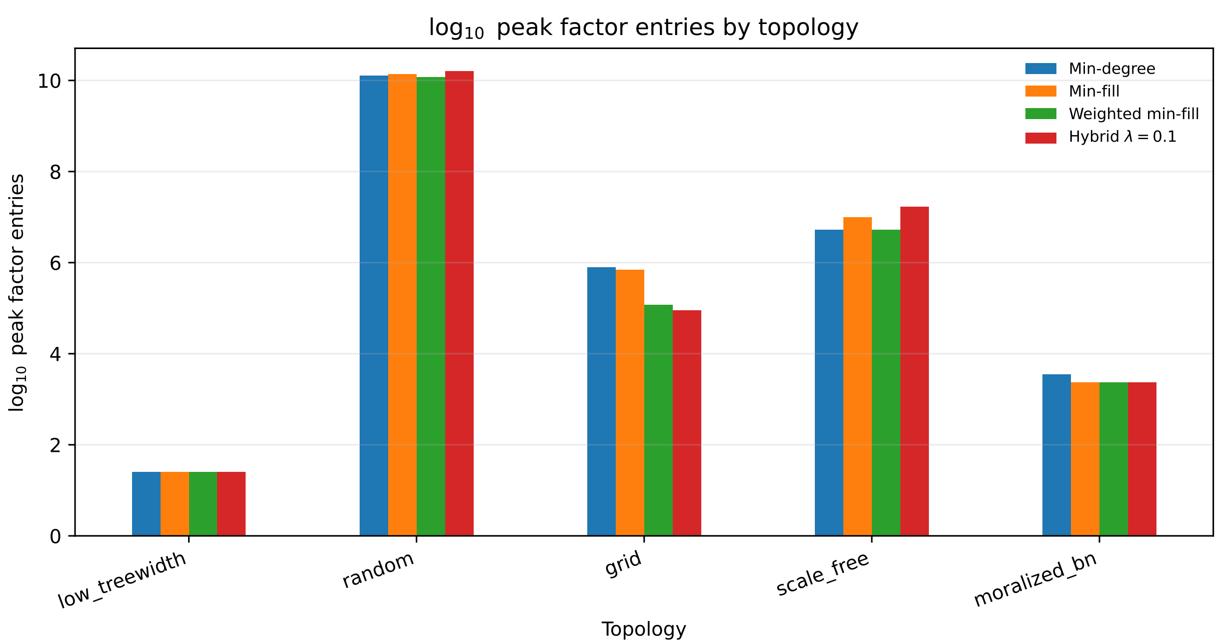
Lambda Sensitivity

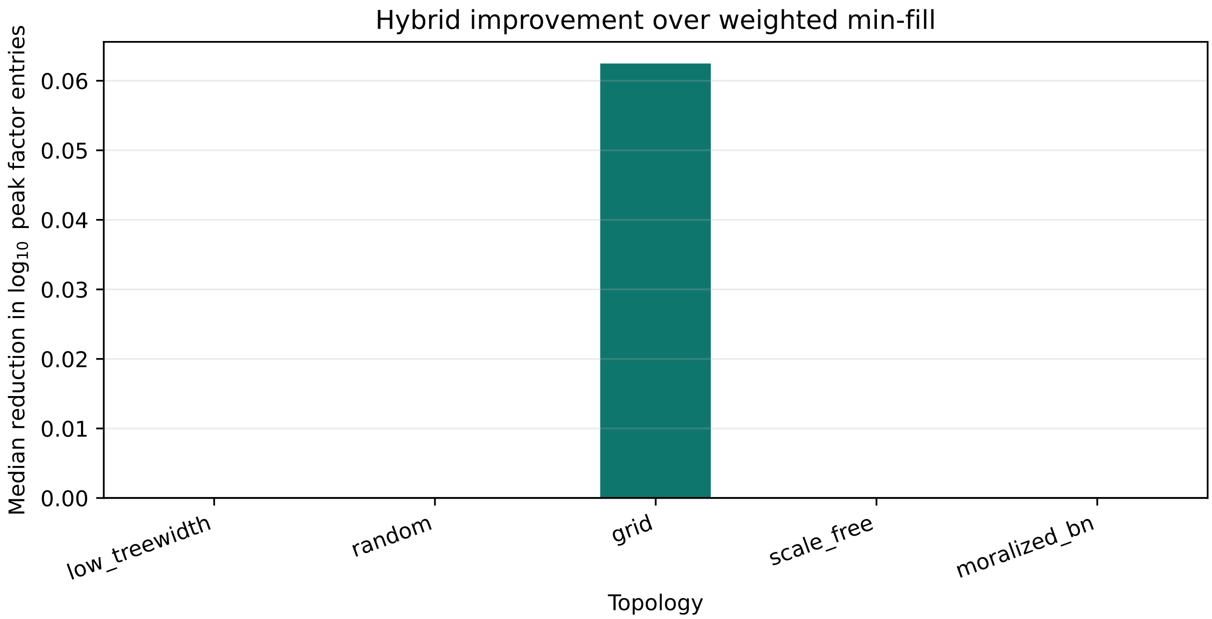
The hybrid method is not uniformly monotone in λ. A fixed $\lambda = 0.1$ was used as the main comparison point because it keeps the downstream-pressure term active without letting it dominate weighted fill cost.

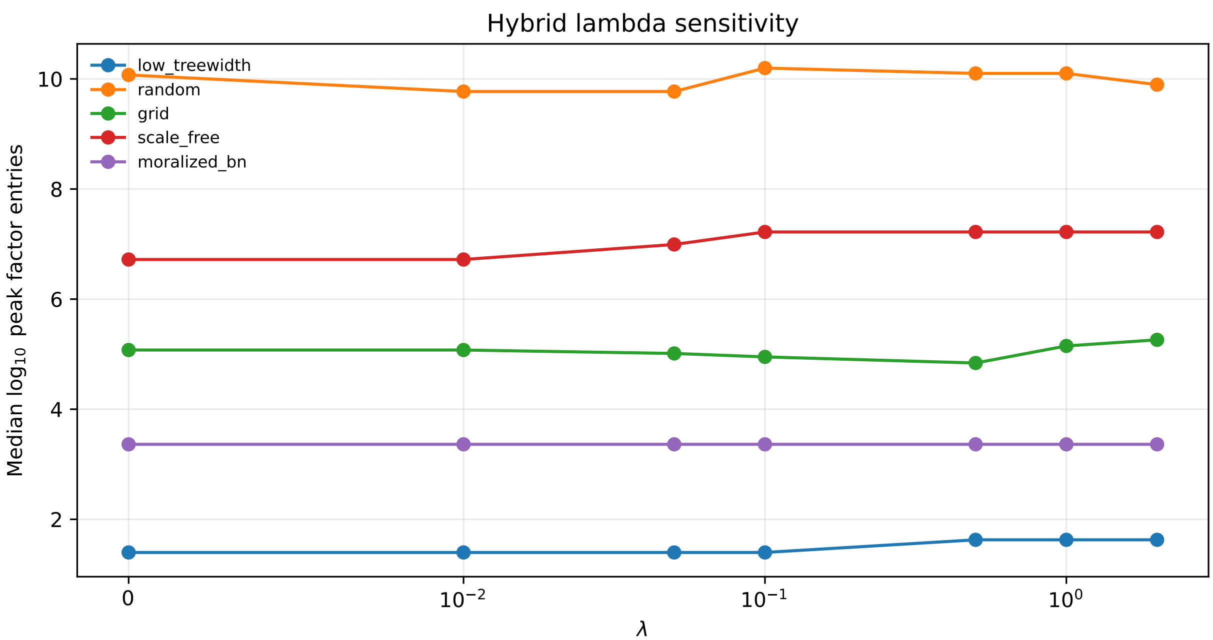
Topology	Best λ	Median $\log_{10}$ peak size
Low-treewidth	0.00	1.40
Random	0.01	9.77
Grid	0.50	4.84
Scale-free	0.00	6.72
Moralized BN	0.00	3.36

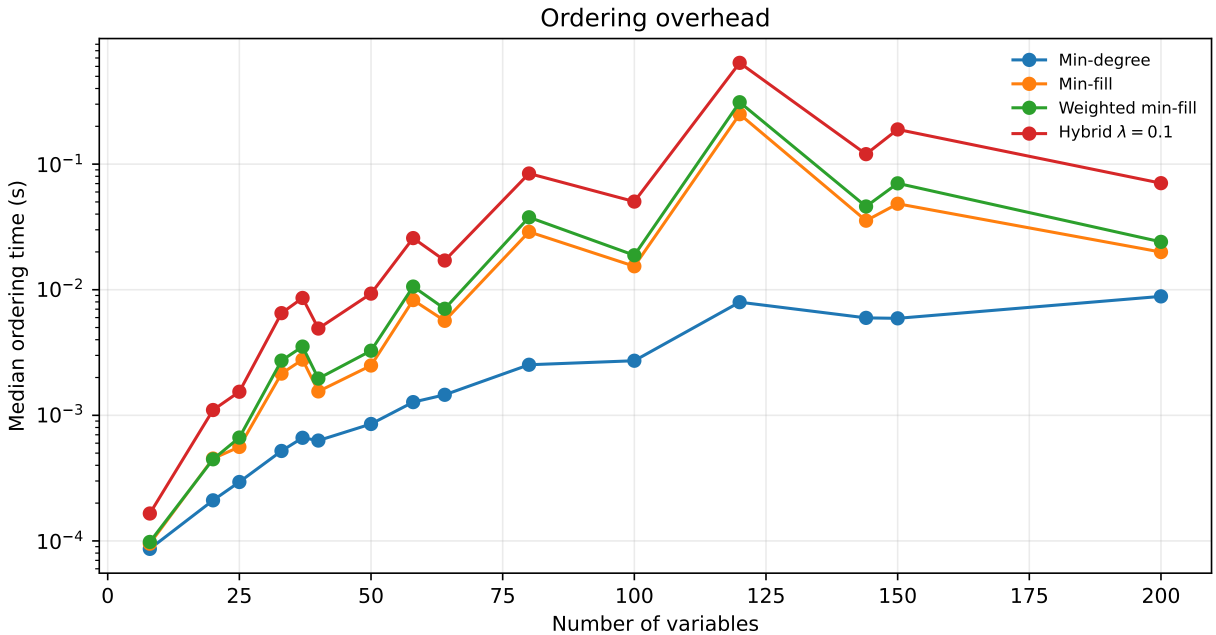
Figures


Induced width by topology
Min-degree
Min-fill
Weighted min-fill
Hybrid $\lambda = 0.1$
Induced width
0
5
10
15
20
low_treewidth
random
grid
scale_free
moralized_bn
Topology


$\log_{10}$ peak factor entries by topology
Min-degree
Min-fill
Weighted min-fill
Hybrid $\lambda = 0.1$
$\log_{10}$ peak factor entries
0
2
4
6
8
10
low_treewidth
random
grid
scale_free
moralized_bn
Topology


Hybrid improvement over weighted min-fill
Median reduction in log10 peak factor entries
0.00
0.01
0.02
0.03
0.04
0.05
0.06
low_treewidth
random
grid
scale_free
moralized_bn
Topology


Hybrid lambda sensitivity
low_treewidth
random
grid
scale_free
moralized_bn
Median log10 peak factor entries
2
4
6
8
10
0
10^{-2}
10^{-1}
10^{0}
λ



Interpretation

- Low-treewidth chains and trees behaved as sanity checks: structural metrics were small and the strongest heuristics were usually close.
- Random graphs became difficult as density increased; peak factor estimates were more informative than raw runtime because many dense instances hit the explicit-VE cutoff.
- Grid graphs showed the expected sensitivity to ordering, with induced width and peak factor estimates separating the methods more clearly as the grid grew.
- Scale-free graphs highlighted the hub risk: methods that create dense neighborhoods around high-degree variables produced larger peak factors.
- The hybrid heuristic helped on some topology groups but did not dominate weighted min-fill uniformly, which is a useful empirical finding. The downstream-pressure proxy can capture some lookahead signal, but its benefit depends on graph structure and the chosen λ.

Bounded Lookahead Check

The bounded-lookahead oracle was run only on small graphs; it produced 5 successful rows. This keeps the comparison computationally feasible while still giving a reference point for how much a more explicit lookahead objective can improve structural metrics.

Limitations

This run is self-contained and reproducible, but the BN benchmark portion should be strengthened in a future pass by loading canonical BIF/UAI networks directly through `pgmpy` or `pyAgrum`. The current report also treats exact VE runtime as secondary evidence because structural complexity, especially peak factor size, is the more stable measure across high-treewidth instances.